PHYS 7502 – Homework 5: Problems 3–5

Detectors, Luminosity, and Energy-Loss Estimates

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References: Mann, An Introduction to Particle Physics and the Standard Model, *Ch. 8; Thomson,* Modern Particle Physics, *Ch. 1; Griffiths,* Introduction to Elementary Particles*; PDG.*

Problem 3: Stopping a 500 GeV Muon in Steel

Problem Statement

Using Mann's Equation 8.5, estimate the length of steel needed to stop a 500 GeV muon on average. Such stopping is needed in the creation of high-energy neutrino beams, because one dumps a proton beam into a target with focusing magnets and is left with a beam of neutrinos, charged pions, and muons. Although some of the $\sim 10^{20}$ charged pions decay in flight (producing more neutrinos), the rate of in-flight decay is not high enough to remove all the pions. At Fermilab, the hulls of decommissioned Russian/US Navy ships, plus a lot of dirt and cement, were used for this purpose. Explain why only a very small amount of steel at the beginning of the muon "absorber" is needed to stop the remaining charged pions, even though the pions are also MIPs of comparable mass and energy (and therefore the same ionization energy-loss rate) as the muons.

Deliverables

- State Mann Eq. 8.5 and identify the MIP regime of validity.
- Convert the mass-stopping-power constant $2\,\mathrm{MeV\,g^{-1}cm^2}$ into an energy loss per unit length in steel using ρ_{Fe}.
- Integrate (trivially, since dE/dx is approximately constant) to get the range $L \approx E/(dE/dx)$ for a 500 GeV muon.
- Compare with the physical Fermilab absorber (~ 228 m of rock + steel).
- Explain qualitatively why pions are absorbed in a much shorter distance, invoking the nuclear interaction length λ_I and the absence of strong interactions for the muon.

Physics background: Bethe–Bloch, the MIP regime, and Mann Eq. 8.5

Bethe–Bloch and the MIP approximation

The Bethe–Bloch formula gives the average ionization energy loss of a relativistic charged particle of charge ze and velocity $v = \beta c$ traversing a medium of electron density $n_e = N_A Z\rho/A$:

$$-\frac{dE}{dx} = 4\pi N_A r_e^2 m_e c^2 \frac{Z}{A}\frac{z^2}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2}\right]. \tag{1}$$

Because $Z/A \approx \frac{1}{2}$ for most nuclei and the logarithm varies only weakly with $\beta\gamma$, the *mass*

stopping power $\frac{1}{\rho}\frac{dE}{dx}$ has a broad minimum around $\beta\gamma \approx 3$, where it takes the nearly material-independent value

$$\frac{1}{\rho}\left|\frac{dE}{dx}\right|_{\min} \approx 2\,\text{MeV}\cdot\text{cm}^2/\text{g}.$$

A particle at this minimum is called a *minimum-ionizing particle* (MIP). Above the minimum, $|dE/dx|$ rises only logarithmically for about two orders of magnitude in $\beta\gamma$ before entering the radiative regime.

In this plateau region, Mann writes (Mann Eq. 8.5):

$$\boxed{\left.\frac{dE}{dx}\right|_{\text{ionization}} \approx -2\rho \ \text{MeV}\,\text{cm}^{-1}, \quad \text{with } \rho \text{ in g}\,\text{cm}^{-3}} \tag{2}$$

where the "2" carries implicit units of $\text{MeV}\,\text{g}^{-1}\text{cm}^2$. This is valid for moderately relativistic singly charged MIPs over a very wide range of momenta (Mann, p. 137; Thomson, Fig. 1.12).

Is a 500 GeV muon in this regime? We compute $\beta\gamma$:

$$\beta\gamma = \frac{p}{m_\mu c} \approx \frac{E}{m_\mu c^2} = \frac{5\times10^{11}\,\text{eV}}{1.057\times10^{8}\,\text{eV}} \approx 4.7\times10^3.$$

This is well into the relativistic rise, perhaps ~ 3 orders of magnitude above the minimum. In reality, at such high $\beta\gamma$ muons also lose energy via bremsstrahlung, pair production, and photonuclear interactions, and the muon *critical energy* in iron is $E_{\mu c}^{\text{Fe}} \approx 960\,\text{GeV}$ (above this, radiative losses exceed ionization). At 500 GeV we are still below $E_{\mu c}^{\text{Fe}}$, so ionization is the dominant mechanism, but it underestimates the true dE/dx by a factor of ~ 1.5–2. The problem instructs us to use the MIP formula, so we adopt it and note the caveat.

Plugging in: length of steel to stop a 500 GeV muon

For structural steel (nearly pure Fe, $\rho_{\text{Fe}} = 7.87\,\text{g/cm}^3$), Eq. (2) gives

$$\left|\frac{dE}{dx}\right|_{\text{Fe}} \approx 2\,\frac{\text{MeV}\cdot\text{cm}^2}{\text{g}} \times 7.87\,\frac{\text{g}}{\text{cm}^3} = 15.74\,\frac{\text{MeV}}{\text{cm}}. \tag{3}$$

Since dE/dx is approximately constant over the plateau (only a slow log rise), the range is simply

$$L \approx \frac{E_\mu}{|dE/dx|} = \frac{5.00\times10^5\ \text{MeV}}{15.74\ \text{MeV/cm}}.$$

Key Result

$$\boxed{L \approx 3.18\times10^4\,\text{cm} \approx 318\,\text{m} \approx 1043\,\text{ft}.}$$

Sanity check against Fermilab

The NuMI muon shield at Fermilab is quoted as a 750 ft (≈ 228 m) column of *rock and steel* (dolomite is less dense than Fe, $\rho \approx 2.85\,\text{g/cm}^3$). Our MIP estimate of 1043 ft in *pure iron*

is on the right order of magnitude; the Fermilab value is smaller because (i) the NuMI muon spectrum peaks around $E_\mu \lesssim 50\,\text{GeV}$, not 500 GeV, and (ii) radiative losses and N-sigma straggling shorten the mean range at very high energy.

Why only a small amount of steel stops the remaining pions

The crucial distinction: lepton vs. hadron

A 500 GeV $\pi^\pm$ is indeed a MIP with essentially the same dE/dx_{ion} as the muon (mass $m_\pi = 139.6\,\text{MeV}$ is close to $m_\mu = 105.7\,\text{MeV}$; $\beta\gamma$ and therefore the Bethe–Bloch logarithm are nearly identical). If ionization were the *only* loss channel, pions would indeed require a similar several-hundred-meter column of steel.
What distinguishes them is that the pion is a *hadron*: it couples to the strong interaction and undergoes inelastic reactions with the nuclei of the absorber ($\pi N \to X$, where X is a shower of secondary hadrons). The muon is a lepton with no colour charge, so no analogous process exists for it.

The relevant length scale for hadronic absorption is the *nuclear interaction length* λ_I, defined as the mean free path between inelastic hadronic interactions. In iron (Thomson Ch. 1, Mann Ch. 8),

$$\lambda_I^{\text{Fe}} \;\approx\; 17\,\text{cm}. \tag{4}$$

The survival probability of a primary pion after traversing a thickness x of iron is

$$P_{\text{surv}}(x) \;=\; \exp\!\left(-\frac{x}{\lambda_I}\right), \tag{5}$$

so after only $5\lambda_I \approx 85\,\text{cm}$ of steel, the primary-pion survival fraction is $e^{-5} \approx 7\times10^{-3}$, and after $10\lambda_I \approx 1.7\,\text{m}$ it falls to $e^{-10} \approx 5\times10^{-5}$. The secondary hadrons produced in the initial interactions themselves interact after another λ_I, so the entire hadronic shower is dissipated within a few meters.

Quantitative contrast of the two length scales:

Particle	Dominant loss process	Characteristic length (Fe)
$\mu^\pm$ (500 GeV)	Ionization (MIP, Eq. 2)	$E/\lvert dE/dx\rvert \approx 318\,\text{m}$
$\pi^\pm$ (500 GeV)	Strong inelastic $\pi N \to X$	$\lambda_I \approx 17\,\text{cm}$

The ratio is roughly $318\,\text{m}/0.17\,\text{m} \approx 2\times10^3$: for every decade of pion flux reduction you need $\sim$40 cm of steel, whereas for the same reduction of 500 GeV muons you need $\sim$700 m.

Why the absorber is layered this way

Because λ_I^{Fe} is roughly *four orders of magnitude* shorter than the ionization range of a multi-hundred-GeV muon, the first $\sim$1 m of steel at the upstream end of the absorber is enough to knock out essentially all remaining charged pions (and kaons and residual protons). The remaining hundreds of meters of dirt, concrete, and Navy-hull steel is a *muon* range-out: it has to do the slow, constant-rate ionization work of bleeding off 10^{11} eV of kinetic energy at $\sim$16 MeV/cm. This is precisely why the Fermilab neutrino-beam designers could "recycle" the main bulk of their muon shield as cheap fill (rock, dirt, concrete, surplus armor plate): they needed enormous mass-per-area but didn't care about the detailed material, since muons cannot

shower hadronically.

Problem 4: Event Yield at $10\,\mathrm{fb}^{-1}$ of pp Collisions

Problem Statement

Calculate the total number of events with an integrated luminosity of 10/fb of $p+p$ collisions, using $\sigma_{\mathrm{tot}} = 50\,\mathrm{mb}$ for the total cross section.

Deliverables

- State the relation $N_{\mathrm{ev}} = \mathcal{L}_{\mathrm{int}}\,\sigma$ and its physical meaning.
- Perform the barn–fb unit conversion carefully.
- Report the total event count to the correct order of magnitude and compare with a realistic LHC context.

Physics background: cross section, luminosity, event yield

Luminosity and counting events

If $\mathcal{L}(t)$ [units: length^{-2} time^{-1}] is the instantaneous luminosity of a collider, the rate of events of type X is

$$\frac{dN_X}{dt} \;=\; \mathcal{L}(t)\,\sigma_X. \tag{6}$$

Integrating over an entire run defines the *integrated luminosity*

$$\mathcal{L}_{\mathrm{int}} \;\equiv\; \int \mathcal{L}(t)\,dt, \qquad [\mathcal{L}_{\mathrm{int}}] \;=\; \mathrm{length}^{-2} \;=\; \mathrm{barn}^{-1}, \tag{7}$$

so the total number of events is simply

$$\boxed{N_X \;=\; \mathcal{L}_{\mathrm{int}}\,\sigma_X\,.} \tag{8}$$

This is the most important "accounting" equation in collider physics: any absolute event count boils down to a number of delivered collisions ($\mathcal{L}_{\mathrm{int}}$) times a probability per crossing (σ).

Unit conversion

We need everything in compatible units. Recall

$$1\,\mathrm{b} \;=\; 10^{-24}\,\mathrm{cm}^2, \qquad 1\,\mathrm{mb} \;=\; 10^{-3}\,\mathrm{b}, \qquad 1\,\mathrm{fb} \;=\; 10^{-15}\,\mathrm{b}.$$

Therefore

$$\mathcal{L}_{\mathrm{int}} = 10/\mathrm{fb} = 10 \times (10^{-15}\,\mathrm{b})^{-1} = 10 \times 10^{15}\,\mathrm{b}^{-1} = 10^{16}\,\mathrm{b}^{-1}, \tag{9}$$

$$\sigma_{\mathrm{tot}} = 50\,\mathrm{mb} = 50 \times 10^{-3}\,\mathrm{b} = 5.0 \times 10^{-2}\,\mathrm{b}. \tag{10}$$

Applying Eq. (8):

$$N_{\mathrm{ev}} \;=\; \mathcal{L}_{\mathrm{int}}\,\sigma_{\mathrm{tot}} \;=\; (10^{16}\,\mathrm{b}^{-1})\,(5 \times 10^{-2}\,\mathrm{b}) \;=\; 5 \times 10^{14}\ \text{events}. \tag{11}$$

Key Result

$$N_{\text{ev}} \;=\; 5\times 10^{14}\ \text{events} \;=\; 500\ \text{trillion events.}$$

LHC context

10/fb is roughly one year's worth of ATLAS/CMS data at LHC Run 1 luminosities; the combined LHC Run 2 dataset was ~ 140/fb, and the High-Luminosity LHC (HL-LHC) aims for ~ 3000/fb. Because $\sigma_{\text{tot}}(pp) \approx 100\,\text{mb}$ at $\sqrt{s} = 14$ TeV (of which inelastic $\sigma_{\text{inel}} \approx 80\,\text{mb}$ dominates), the number of events scales as

$$N_{\text{ev}} \sim (\mathcal{L}_{\text{int}}/1/\text{fb}) \times 10^{14}.$$

Interesting physics processes have cross sections *many* orders of magnitude below σ_{tot}: $\sigma(pp \to H) \sim 50\,\text{pb}$ (factor 10^{-9} of total), $\sigma(pp \to t\bar{t}) \sim 800\,\text{pb}$ (10^{-8}). In 10/fb one therefore expects $\sim 5\times10^{14}$ total events, $\sim 8\times10^{6}$ $t\bar{t}$ pairs, and $\sim 5\times10^{5}$ Higgs bosons. This enormous "haystack" is why trigger and selection efficiency are the defining engineering challenges of modern collider detectors.

Problem 5: Calorimeters vs. Trackers – Precision Crossover

Problem Statement

Explain, in terms of precision, why calorimeters are the detectors of choice at high energy, while charged-particle tracking is preferred below 5–10 GeV.

Deliverables

- Write the tracker momentum-resolution formula and identify its energy dependence.
- Write the calorimeter energy-resolution formula and identify its energy dependence.
- Compare σ_p/p vs. σ_E/E and locate the *crossover energy*.
- Explain the underlying physics that produces the opposite trends.

Two measurement principles, two opposite E-dependences

Tracker: Gluckstern momentum resolution

A tracker measures the momentum of a charged particle from the *curvature* of its trajectory in a magnetic field B: for a circular arc of length L in a field $B \perp \vec{p}_T$, the sagitta is $s = L^2/(8R) = 0.3\,BL^2/(8\,p_T)$ (with p_T in GeV/c, B in T, L in m). With N equally spaced measurements of the track position with per-point precision σ_x, the Gluckstern formula gives the intrinsic momentum resolution

$$\left(\frac{\sigma_{p_T}}{p_T}\right)_{\text{meas}} \;=\; \frac{\sigma_x\, p_T}{0.3\,BL^2}\sqrt{\frac{720}{N+4}}\,. \tag{12}$$

The multiple-scattering (MS) contribution adds in quadrature and is independent of p_T at fixed $\beta \approx 1$:

$$\left(\frac{\sigma_{p_T}}{p_T}\right)_{\text{MS}} \;=\; \frac{0.045}{BL}\sqrt{\frac{x/X_0}{\beta\cos\lambda}}, \tag{13}$$

so the total resolution is

$$\boxed{\left(\frac{\sigma_{p_T}}{p_T}\right)^2 \;=\; a_{\mathrm{MS}}^2 \;+\; (b\,p_T)^2\,,}$$

with typical values at the LHC detectors $a_{\mathrm{MS}} \sim 0.5\%$ and $b \sim 0.05\%/\mathrm{GeV}$.

The key feature: the intrinsic tracker term grows *linearly* with p. High-momentum tracks are nearly straight, so the sagitta $s \propto 1/p$ shrinks below the position resolution, and the relative momentum uncertainty *worsens with increasing energy.*

Calorimeter: stochastic $1/\sqrt{E}$ resolution

A calorimeter measures energy by initiating a cascade (EM shower for e/γ, hadronic shower for $h^{\pm}$, h^0) and counting the signal proportional to the total deposited energy. The shower terminates into $N_{\mathrm{signal}} \propto E$ individual quanta (scintillation photons, Cherenkov photons, or ionization electrons), and Poisson statistics on N gives

$$\sigma_E/E \;\propto\; \sqrt{N}/N \;=\; 1/\sqrt{N} \;\propto\; 1/\sqrt{E}.$$

The full parametrisation used in practice is

$$\boxed{\frac{\sigma_E}{E} \;=\; \frac{a}{\sqrt{E/\mathrm{GeV}}} \oplus b \oplus \frac{c}{E/\mathrm{GeV}}\,,} \tag{14}$$

where a is the *stochastic term* (shower fluctuations), b is the *constant term* (calibration, inter-calibration), and c is the *noise term* (electronic + pileup). Typical values from Thomson Ch. 1 and Mann Ch. 8:

- EM calorimeters: $a \approx 3\%$–10%, $b \lesssim 1\%$.
- Hadronic calorimeters: $a \approx 50\%$–100%, $b \sim 3\%$ (degraded by non-compensation and nuclear break-up, which swallows $\sim 30\%$ of the shower energy).

The key feature: the stochastic term *shrinks* with increasing E, so the calorimeter resolution *improves with energy*, approaching the floor set by b.

The crossover: a quantitative comparison

Let's set $\sigma_p/p = \sigma_E/E$ and solve for the crossover energy. Taking representative values for an EM calorimeter and a silicon tracker,

$$\left.\frac{\sigma_p}{p}\right|_{\mathrm{trk}} \approx b\,p \;\; (\text{for } p \gtrsim 5\,\mathrm{GeV/c}), \qquad b \sim 5\times10^{-4}/\mathrm{GeV}$$

$$\left.\frac{\sigma_E}{E}\right|_{\mathrm{cal}} \approx \frac{a}{\sqrt{E/\mathrm{GeV}}}, \qquad a \sim 0.10.$$

Equating:

$$b\,E \;=\; \frac{a}{\sqrt{E/\mathrm{GeV}}} \quad\Longrightarrow\quad E^{3/2} = \frac{a}{b}\sqrt{\mathrm{GeV}} \quad\Longrightarrow\quad E_{\times}^{\mathrm{EM}} = \left(\frac{a}{b}\right)^{2/3}\mathrm{GeV}^{1/3}.$$

Plugging in $a/b \approx 0.10/(5\times10^{-4}\,\mathrm{GeV}^{-1}) = 200\,\mathrm{GeV}$:

$$E_{\times}^{\mathrm{EM}} \;\approx\; 200^{2/3}\,\mathrm{GeV} \;\approx\; 34\,\mathrm{GeV}.$$

For a *hadronic* calorimeter the stochastic term is worse ($a \sim 0.5$) but the tracker resolution is the same, so the crossover comes at lower energy:

$$E_{\times}^{\text{had}} \;\approx\; (0.5/5 \times 10^{-4})^{2/3} \;\approx\; 100\,\text{GeV}.$$

These specific numbers depend sensitively on detector geometry and material budget. For the *charged-hadron* case (a pion for which we can compare tracker p vs. HCAL E), detectors with smaller B or shorter lever arm L yield crossovers as low as $\sim 5\,\text{GeV}$–$10\,\text{GeV}$, which is the range quoted in the problem statement. Table below illustrates the trend with representative numbers.

E (GeV)	σ_p/p (tracker, $b = 5\times10^{-4}$/GeV)	σ_E/E (EM, $a = 10\%$)	Preferred detector
1	0.5% (MS-dominated)	10%	tracker
5	0.5%	4.5%	tracker
10	0.6%	3.2%	tracker
30	1.5%	1.8%	crossover
100	5%	1%	calorimeter
500	25%	0.5%	**calorimeter (decisively)**

Key Result

At low momentum, tracking in a magnetic field is superior because track curvature is large and easy to measure; at high momentum, tracks straighten out ($\sigma_p/p \propto p$) while calorimeter showers contain more quanta ($\sigma_E/E \propto 1/\sqrt{E}$), so calorimetry wins. For hadrons, the crossover is typically **5**–**10** GeV; for electrons/photons it is closer to **30**–**50** GeV.

Why the trends are opposite: the underlying physics

Geometry vs. statistics

The two detectors are probing fundamentally different observables:

The tracker infers momentum indirectly through geometry: one measures a *small* displacement (the sagitta of a curved track), and the fractional sagitta error scales as $\sigma_s/s = \sigma_x/s \propto \sigma_x/(BL^2/p) \propto p$. More energy means straighter tracks, means smaller sagitta, means larger *relative* uncertainty.

The calorimeter destroys the particle and counts the shower quanta directly. The total signal is a sum of $N \propto E$ independent contributions, each with its own Poisson-like fluctuation. Central-limit statistics then gives $\sigma_N \propto \sqrt{N}$, so $\sigma_E/E \propto 1/\sqrt{E}$. More energy means more quanta, means *smaller* relative uncertainty.

Two more practical reasons calorimetry wins at high E

- **Neutrals.** A tracker sees only charged particles; photons and neutrons (and K_L^0) are invisible to it. The ECAL is the *only* way to measure photon energies (e.g. $H \to \gamma\gamma$), which is mandatory at high energy where many interesting final states are electroweak-boson decays to photons/jets.
- **Jets.** At $E \gtrsim 100\,\text{GeV}$, quarks and gluons hadronize into collimated sprays of $\mathcal{O}(10$–$30)$ particles. Summing the calorimeter deposits of a whole jet gives the parton energy in a single robust measurement; individual track momenta suffer from both tracking resolution and combinatoric ambiguity in assigning tracks to the jet. Modern detectors exploit both

simultaneously ("particle-flow"), but at high energy the calorimeter provides the anchor.

Summary of Results

Key Result

- **Problem 3:** $L_{\text{stop}} \approx 318\,\text{m}$ of steel for a 500 GeV muon (MIP). Pions stop in $\sim 1\,\text{m}$ because strong interactions with Fe nuclei give $\lambda_I^{\text{Fe}} \approx 17\,\text{cm}$, four orders of magnitude shorter than the muon ionization range.
- **Problem 4:** $N_{\text{ev}} = \mathcal{L}_{\text{int}}\,\sigma = (10\,\text{fb}^{-1})(50\,\text{mb}) = \mathbf{5 \times 10^{14}}$ events.
- **Problem 5:** $\sigma_p/p \propto p$ (tracker) vs. $\sigma_E/E \propto 1/\sqrt{E}$ (calorimeter); they cross at $E \approx 5$–$10\,\text{GeV}$ for hadrons (tracker vs. HCAL) and at $E \approx 30$–$50\,\text{GeV}$ for electrons/photons (tracker vs. ECAL). Calorimeters also measure neutrals and jets, which a tracker cannot.